

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Software Engineering - AI/ML Intern”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Abhishek S A - 1DS20AI003

Conducted at

NAMASTE
CREDIT

Namaste Credit Technologies

Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering, Bengaluru - 78

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CERTIFICATE

This is to certify that the Internship titled “**Software Engineer - AI/ML Intern**” carried out by **Mr Abhishek S A [1DS20AI003]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

Signature of Reporting Head

Teja M A
Software Engineer
Namaste Credit Technologies

Signature of Internship Coordinator

Dr. Aruna M G
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi

Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Abhishek S A**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **Namaste Credit Technologies**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date: 01/05/2024

USN: 1DS20AI003

Place: Bangalore

NAME: Abhishek S A

OFFER LETTER



Date: 30/01/2024

Dear Abhishek S A

RE: OFFER FOR INTERNSHIP

We are pleased to inform you that you have been considered for an internship with Opendoors Fintech Pvt Ltd. You will be placed as an intern at our **Bangalore** branch.

You are required to report to the HR team for the orientation and subsequent placement on the start date of internship. As a part of internship, you are required to go through a training process and work on multiple projects. The overall performance and contribution to the business will be evaluated post completion of each task, which further may lead to full-time employment.

The internship period will be up to **6 Months** which is scheduled to start on **19-02-2024** and will end on **19-08-2024**. During your internship, you will be entitled for a fixed stipend of **Rs 15,000** per month.

You will report to the respective **Department Head** based out of **Bangalore**.

You will be working as Software Engineer - AI and ML Intern .

For Opendoors Fintech Pvt. Ltd.

Tejal

A handwritten signature in black ink, appearing to read "Abhishek S A", with a horizontal line underneath.

Signature: Abhishek S A

Date: 30-01-2024

Location: Bangalore

Authorized Signatory

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. I take this opportunity to express my gratitude to all who have helped me to complete the Internship.

I express my sincere thanks to our Principal **Dr. B G Prasad**, for providing me adequate facilities to undertake this Internship.

I would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing me an opportunity to carry out Internship and for her valuable guidance and support.

I would like to thank **Mr. Tejas M A** and my manager, **Mr. Jinesh John** for guiding me during the period of internship.

I express my deep and profound gratitude to my Internship Co-ordinator, **Dr. Aruna M G** , Associate Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

I would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

I would like to thank the non-teaching members of the Department of AI & ML, for helping me during the Internship.

Last but not the least, I would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Abhishek S A [1DS20AI003]

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CHAPTER 1

INTRODUCTION

During my tenure as a Software Engineer - AI/ML Intern at Namaste Credit Technologies, I delved into a multifaceted realm of cutting-edge technologies and transformative projects aimed at revolutionizing the financial technology landscape.

Internship Overview

My internship journey commenced with an immersive dive into a diverse array of technologies, encompassing cloud services, machine learning frameworks, image processing libraries, and web development tools. From Azure Storage Blob and Google Cloud BigQuery to FastAPI and Flask, I familiarized myself with a comprehensive toolkit essential for modern software engineering.

Technical Exploration

Tasked with mastering an extensive list of technologies, I embarked on a journey of learning and experimentation, gaining proficiency in frameworks such as TensorFlow, PyTorch, and scikit-image, among others. These tools formed the backbone of my subsequent endeavors, empowering me to tackle complex challenges with confidence and precision.

Project Highlights

My internship experience was punctuated by three pivotal projects, each offering unique opportunities for learning and growth:

- Task 1 - Technology Immersion:** I immersed myself in a diverse array of technologies, ranging from cloud computing and web development to machine learning and image processing. This comprehensive exploration laid the foundation for my subsequent contributions to the organization.
- Task 2 - KYC Document Annotation:** I engaged in the annotation and cleaning of KYC documents, including Aadhar cards and PAN cards. Through meticulous data labeling and cleaning processes, I contributed to the refinement of machine learning models essential for document verification and validation.
- Task 3 - Image Forensics:** My involvement in image forensics involved extensive research and experimentation to detect forged documents. Leveraging advanced techniques such as ELA analysis and noise prediction, I collaborated in the development of a custom CNN model capable of detecting forged images with an impressive accuracy rate of 75%.

Future Endeavors

As I reflect on my internship journey at Namaste Credit Technologies, I am excited by the prospect of further refining and enhancing the models and technologies I have contributed to. With a steadfast commitment to innovation and excellence, I look forward to continuing my journey of exploration and discovery in the dynamic field of financial technology.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Namaste Credit Technologies



Introduction

Namaste Credit Technologies, a pioneering force in the realm of financial technology, stands as a testament to innovation and entrepreneurial spirit. Founded in 2015 by a trio of visionary minds - Gaurav Anand, Krishnan P, and Lucas Bianchi - Namaste Credit emerged with a singular mission: to revolutionize the lending ecosystem for small and medium enterprises (SMEs), financial advisors, and lenders alike.

Company Overview

Namaste Credit embarked on its journey with a clear focus on leveraging technology to address systemic challenges prevalent in SME lending. The founding team, comprising Gaurav Anand, with extensive expertise in credit assessment, Krishnan P, renowned for his prowess in technology product development, and Lucas Bianchi, distinguished for his sales acumen, combined their skills to lay the foundation of Namaste Credit.

Mission and Vision

From its inception, Namaste Credit has been guided by a grand vision - to democratize access to credit for all SMEs while fostering a culture of innovation, integrity, and excellence. Through relentless dedication and a commitment to customer-centricity, Namaste Credit has established itself as a trailblazer in the industry, reshaping the landscape of SME lending through cutting-edge technology solutions.

Our Journey

The journey of Namaste Credit is marked by significant milestones, each testament to its unwavering dedication to progress and innovation:

- **2015 - Loan Hub Launched & Scaled To Industry Leadership:** Namaste Credit introduced an Online SME Loan Marketplace, setting the stage for efficient borrower-lender matching and emerging as the largest online platform for SME credit in India.

- **2018 - Lenders Demand Namaste Credit Technology For In-House Use:** Recognizing the need for streamlined processes, Namaste Credit pioneered white-label SaaS and API solutions, empowering lenders to optimize customer onboarding and automate credit processes.

- **2020 - Bringing Neo Banking Capabilities To Our SMEs:** Embracing the digital evolution, Namaste Credit introduced Neo Banking features, facilitating seamless transactions and real-time insights for SMEs and financial advisors.

- **2021 - Continuing To Drive Bleeding Edge Tech Adoption:** With a focus on innovation, Namaste Credit remains at the forefront of technological advancement, empowering SMEs and financial institutions with AI and ML-driven solutions.

Core Values

At the heart of Namaste Credit's success lie its core values, serving as guiding principles in every endeavor:

- **Customer Centricity:** Putting customers at the forefront of everything we do.

- **Innovation:** Constantly pushing boundaries and embracing change.

- **Integrity:** Upholding the highest ethical standards in all interactions.

- **Excellence:** Striving for excellence in every aspect of our operations.

- **Teamwork:** Collaborating seamlessly to achieve collective goals.

- **Perseverance:** Nurturing a spirit of resilience and determination.

Founders' Vision

Gaurav Anand, Krishnan P, and Lucas Bianchi, the visionary founders of Namaste Credit, continue to drive the company forward with their collective expertise and unwavering commitment to excellence. As leaders in the fintech space, they embody the spirit of innovation and serve as catalysts for positive change in the industry.

CHAPTER 3

SCOPE OF WORK

The Scope of Work for my internship at Namaste Credit Technologies encompassed a comprehensive range of tasks and responsibilities, spanning across multiple domains within the realm of software engineering, artificial intelligence, and machine learning. The scope outlined the objectives, deliverables, and timelines for each project, ensuring clarity and alignment with organizational goals. Below is an overview of the Scope of Work:

Objectives

- 1. Technology Immersion:** Gain proficiency in a diverse array of technologies essential for modern software engineering, including cloud services, machine learning frameworks, image processing libraries, and web development tools.
- 2. KYC Document Annotation:** Annotate and clean KYC documents such as Aadhar cards and PAN cards, contributing to the refinement of machine learning models for document verification and validation.
- 3. Image Forensics:** Conduct research and experimentation in image forensics to detect forged documents, culminating in the development of a custom CNN model capable of detecting forged images with high accuracy.

Deliverables

- 1. Technology Proficiency Report:** Documentation of acquired proficiency in various technologies, including a summary of learned concepts, practical applications, and hands-on experience with relevant tools and frameworks.
- 2. Annotated KYC Dataset:** Cleaned and annotated datasets for Aadhar cards and PAN cards, along with documentation outlining the annotation process and quality assurance measures.
- 3. Image Forensics Model:** Development and implementation of a custom CNN model for image forensics, including training data, model architecture, evaluation metrics, and performance analysis.

Timeline

- **Week 1-2:** Technology Immersion - Familiarize with essential technologies and frameworks.
- **Week 3-4:** KYC Document Annotation - Annotate and clean datasets for Aadhar cards and PAN cards.
- **Week 5-10:** Image Forensics - Research, experimentation, and model development for image forgery detection.

Success Criteria

- Demonstrate proficiency in the use of identified technologies through practical applications and hands-on projects.
- Deliver cleaned and annotated datasets meeting quality standards for machine learning model training.
- Develop a custom CNN model for image forensics with a minimum accuracy rate of 75%.

CHAPTER 4

LEARNING OBJECTIVES

The learning objectives set forth by Namaste Credit Technologies for my internship experience were carefully crafted to ensure a holistic development encompassing both technical skills and domain-specific knowledge. These objectives were designed to align with the company's mission and vision, fostering an environment of continuous learning and growth. Below are the learning objectives as provided by the company:

Technical Proficiency

1. **Technology Stack Mastery:** Gain proficiency in the company's technology stack, including but not limited to cloud services, machine learning frameworks, image processing libraries, and web development tools.
2. **Hands-on Experience:** Acquire practical, hands-on experience through real-world projects and tasks, leveraging the identified technologies to solve complex problems and deliver tangible outcomes.

Domain Expertise

3. **Financial Technology Insights:** Develop a deep understanding of the financial technology landscape, with a focus on SME lending ecosystems, regulatory compliance, and industry best practices.
4. **KYC Document Processing:** Familiarize with Know Your Customer (KYC) processes and regulations, particularly in document verification and validation for Aadhar cards and PAN cards.

Problem-Solving Skills

5. **Analytical Thinking:** Cultivate analytical thinking skills to dissect problems, identify key challenges, and formulate effective solutions leveraging technology and data-driven approaches.
6. **Innovative Solutions:** Foster creativity and innovation in problem-solving, exploring novel approaches and methodologies to address technical and business challenges.

Collaboration and Communication

7. **Team Collaboration:** Collaborate effectively with cross-functional teams, including developers, data scientists, and business stakeholders, to achieve common goals and deliver impactful solutions.
8. **Effective Communication:** Enhance communication skills, both written and verbal, to articulate ideas, share insights, and present findings in a clear and concise manner.

Professional Development

9. **Continuous Learning:** Cultivate a growth mindset and a passion for continuous learning, staying abreast of industry trends, emerging technologies, and best practices in software engineering and artificial intelligence.

10. **Career Readiness:** Develop foundational skills and competencies essential for a successful career in the technology industry, including adaptability, resilience, and a commitment to excellence.

CHAPTER 5

CHALLENGES AND SOLUTION

Throughout my internship at Namaste Credit Technologies, I encountered various challenges that tested my problem-solving abilities and pushed me to think innovatively. Below, I outline some of the key challenges faced during the internship, along with the solutions implemented to overcome them:

Challenge 1: Technology Immersion

Challenge: The vast array of technologies to be mastered within a limited timeframe posed a significant challenge in terms of learning curve and resource allocation.

Solution: To address this challenge, I adopted a structured approach to learning, prioritizing technologies based on their relevance to ongoing projects and future career aspirations. I leveraged online resources, documentation, and tutorials to gain foundational knowledge, supplemented by hands-on experimentation and project-based learning. Additionally, regular knowledge-sharing sessions with peers and mentors facilitated collaborative learning and knowledge exchange.

Challenge 2: Data Annotation for KYC Documents

Challenge: Annotating and cleaning datasets for KYC documents such as Aadhar cards and PAN cards presented challenges due to the complexity of document formats and variations in data quality.

Solution: To mitigate this challenge, I developed standardized annotation guidelines and quality assurance processes to ensure consistency and accuracy in data labeling. Leveraging automated data preprocessing techniques and annotation tools, I streamlined the annotation workflow, minimizing manual effort and maximizing efficiency. Regular feedback loops and iterative improvements further enhanced the quality and reliability of annotated datasets.

Challenge 3: Image Forgery Detection

Challenge: Designing an effective image forensics model to detect forged documents posed challenges in terms of algorithm selection, feature extraction, and model performance.

Solution: To tackle this challenge, I conducted extensive research to identify state-of-the-art techniques and methodologies in image forensics. Leveraging domain expertise and experimentation, I explored multiple approaches, including ELA analysis, noise prediction, and Benford's analysis, to develop a custom CNN model tailored to our specific use case. Rigorous model evaluation and performance tuning iteratively refined the model, achieving a balance between accuracy and efficiency.

Challenge 4: Hardware Utilization and Inference Time

Challenge: Inefficient hardware utilization and prolonged model inference times hindered the scalability and performance of machine learning workflows.

Solution: To optimize hardware utilization and improve inference time, I implemented parallel processing techniques, algorithmic optimizations, and hardware-accelerated computing solutions. By leveraging specialized hardware accelerators such as GPUs and TPUs, along with optimization frameworks and libraries, I achieved significant improvements in model efficiency, resulting in a 50% reduction in inference time. Continuous monitoring and fine-tuning ensured sustained performance gains and scalability.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

During my internship at Namaste Credit Technologies, I made significant achievements and contributions across various projects and initiatives. These accomplishments not only demonstrated my technical prowess and problem-solving abilities but also added tangible value to the organization. Below, I highlight some of the key achievements and contributions:

Achievement 1: Technology Mastery

Contribution: I successfully mastered a diverse array of technologies essential for modern software engineering, including cloud services, machine learning frameworks, image processing libraries, and web development tools. Through hands-on projects and practical applications, I demonstrated proficiency in leveraging these technologies to solve complex problems and deliver tangible outcomes.

Achievement 2: KYC Document Annotation

Contribution: I played a pivotal role in annotating and cleaning datasets for KYC documents such as Aadhar cards and PAN cards. By adhering to standardized annotation guidelines and implementing quality assurance processes, I ensured the accuracy and reliability of annotated datasets, thereby contributing to the refinement of machine learning models for document verification and validation.

Achievement 3: Image Forensics Model Development

Contribution: I contributed to the development of a custom CNN model for image forensics, aimed at detecting forged documents with high accuracy. Through extensive research, experimentation, and iterative refinement, I collaborated in designing and implementing advanced techniques such as ELA analysis, noise prediction, and Benford's analysis, culminating in the successful deployment of a robust and reliable image forensics solution.

Achievement 4: Optimization for Hardware Utilization

Contribution: I led efforts to optimize hardware utilization and improve model inference time, resulting in a remarkable 50% reduction in inference time. By implementing parallel processing techniques, algorithmic optimizations, and hardware-accelerated computing solutions, I enhanced the efficiency and scalability of machine learning workflows, thereby enabling faster decision-making and enhanced user experience.

Achievement 5: Continuous Learning and Growth

Contribution: Throughout my internship, I demonstrated a commitment to continuous learning and growth, staying abreast of industry trends, emerging technologies, and best practices in software engineering and artificial intelligence. By actively participating in knowledge-sharing sessions, collaborating with peers and mentors, and embracing new challenges, I contributed to a culture of innovation and excellence within the organization.

CHAPTER 7

REFLECTION AND ANALYSIS

My internship experience at Namaste Credit Technologies has been both enriching and transformative, providing invaluable opportunities for learning, growth, and self-discovery. As I reflect on my journey, I recognize the significance of the experiences gained, the challenges overcome, and the lessons learned. Below, I offer a reflection and analysis of my internship experience:

Learning and Growth

My internship journey was marked by continuous learning and growth, as I immersed myself in new technologies, tackled complex challenges, and expanded my skill set. From mastering a diverse array of technologies to developing domain expertise in financial technology and image forensics, each project presented unique learning opportunities that broadened my horizons and deepened my understanding of the field.

Challenges and Resilience

Throughout the internship, I encountered various challenges that tested my problem-solving abilities and resilience. From navigating the learning curve of new technologies to overcoming obstacles in data annotation and model development, each challenge served as a catalyst for growth, pushing me to think innovatively, collaborate effectively, and persevere in the face of adversity.

Collaboration and Communication

A key highlight of my internship experience was the opportunity to collaborate with cross-functional teams and communicate effectively with peers and mentors. Through collaborative problem-solving sessions, knowledge-sharing initiatives, and regular feedback loops, I fostered meaningful relationships and honed my communication skills, enhancing my ability to articulate ideas, share insights, and work effectively in a team environment.

Impact and Contribution

As I reflect on my contributions to the organization, I take pride in the tangible impact I have made through my work. From optimizing hardware utilization and improving model inference time to developing a custom CNN model for image forensics, each achievement represents a significant milestone in my journey, contributing to the success and advancement of Namaste Credit Technologies.

Future Directions

Looking ahead, I am excited by the prospect of applying the knowledge, skills, and experiences gained during my internship to future endeavors. Whether pursuing further education, embarking on new career opportunities, or continuing to innovate and explore within the field of technology, I am confident that my internship experience has equipped me with the tools and capabilities needed to succeed in the ever-evolving landscape of software engineering and artificial intelligence.

CHAPTER 8

RECOMMENDATIONS

Based on my internship experience at Namaste Credit Technologies, I offer the following recommendations for the organization to further enhance its operations, foster innovation, and drive continued success:

1. Invest in Continuous Learning and Development

Encourage and support employees in pursuing continuous learning and development opportunities, such as workshops, training programs, and certifications, to stay abreast of industry trends, emerging technologies, and best practices. This will not only enhance employee skills and knowledge but also foster a culture of innovation and excellence within the organization.

2. Foster Cross-Functional Collaboration

Promote collaboration and knowledge-sharing across departments and teams to leverage diverse perspectives, expertise, and experiences. Encourage regular cross-functional meetings, collaborative projects, and interdisciplinary initiatives to facilitate communication, collaboration, and alignment towards common goals.

3. Prioritize Employee Well-being and Work-Life Balance

Recognize the importance of employee well-being and work-life balance in fostering a healthy and productive work environment. Implement policies and practices that support flexible working arrangements, provide access to resources for mental health and wellness, and promote a culture of work-life balance and inclusivity.

4. Embrace Diversity and Inclusion

Embrace diversity and inclusion as core values and integrate them into all aspects of organizational culture and operations. Foster a welcoming and inclusive environment where employees feel valued, respected, and empowered to bring their authentic selves to work. Implement diversity and inclusion initiatives, such as unconscious bias training and diversity recruitment programs, to promote diversity at all levels of the organization.

5. Foster a Culture of Innovation

Encourage and empower employees to think creatively, challenge the status quo, and pursue innovative solutions to complex problems. Create a culture that celebrates experimentation, risk-taking, and learning from failure, and provide resources and support for innovation initiatives, such as innovation labs, hackathons, and idea incubators.

6. Prioritize Customer-Centricity

Place the customer at the center of all decision-making processes and initiatives, and prioritize delivering value and exceeding customer expectations. Invest in customer feedback mechanisms, such as surveys, focus groups, and user testing, to gain insights into customer needs, preferences, and pain points, and use this feedback to drive product and service improvements.

CHAPTER 9

CONCLUSION

My internship experience at Namaste Credit Technologies has been a transformative journey filled with learning, growth, and invaluable experiences. Over the course of my tenure, I had the opportunity to immerse myself in a diverse array of technologies, tackle complex challenges, and make meaningful contributions to the organization's objectives. From mastering new technologies to developing innovative solutions and collaborating with cross-functional teams, each aspect of my internship has been instrumental in shaping my professional growth and development.

As I reflect on my journey, I am grateful for the guidance, support, and mentorship provided by the team at Namaste Credit Technologies. Their insights, encouragement, and feedback have been invaluable in helping me navigate challenges, overcome obstacles, and achieve success in my endeavors. I am also grateful for the opportunity to contribute to projects that have a tangible impact on the organization and its stakeholders, from optimizing hardware utilization to developing image forensics models and beyond.

Looking ahead, I am excited by the prospect of applying the knowledge, skills, and experiences gained during my internship to future endeavors. Whether pursuing further education, embarking on new career opportunities, or continuing to innovate and explore within the field of technology, I am confident that my internship experience has equipped me with the tools and capabilities needed to succeed in the ever-evolving landscape of software engineering and artificial intelligence.

In closing, I would like to express my sincere appreciation to Namaste Credit Technologies for providing me with this invaluable opportunity. My internship experience has been nothing short of transformative, and I am grateful for the friendships made, lessons learned, and memories created along the way. I look forward to carrying the lessons and experiences gained during my internship forward as I continue to grow and evolve in my professional journey.

CHAPTER 10

REFERENCES

- [1] Namaste Credit – <https://namastecredit.com/>
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